What does conservative form mean?

$$\frac{\partial}{\partial t}\int \rho \vec{v}\, dV$$

↳ There is no derivative outside i.e. nothing outside derivative here: $\frac{\partial}{\partial t}(m) + \frac{\partial}{\partial x}$ } No derivative outside the outermost derivative

$$(x, y, t) \Longleftrightarrow (\xi, \eta, \tau)$$

$$\frac{\partial}{\partial \tau} = \frac{\partial}{\partial t}$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \tau}\frac{\partial \tau}{\partial x} + \cdots$$

$$\frac{\partial^2 u}{\partial y^2} = \left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)\left(\frac{\partial u}{\partial y}\right)$$

$$\left(\xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta}\right)$$

} This is usually sufficient to express in terms of chain rule

↓

strong conservative ($\times J$)

$$J\xi_y \frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial \xi}\left[J\xi_y \frac{\partial u}{\partial y}\right] - \frac{\partial u}{\partial y}\frac{\partial}{\partial \xi}(J\xi_y) \quad \checkmark$$

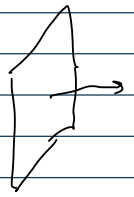
↳ How to write geometric conservation laws

↓

There are 4 general ones → $\frac{\partial}{\partial \xi}\xi_x J + \frac{\partial}{\partial \eta}(\eta_x J) = 0$

If grid is not deformed → $\frac{\partial J}{\partial \tau} = 0$ & $\frac{\partial}{\partial t} = \frac{\partial}{\partial \tau}$

↳ Metric coefficients → (has a physical meaning)



$\vec{A}_\xi = J \nabla \xi$

$= J\xi_x \vec{i} + J\xi_y \vec{j} + J\xi_z \vec{k}$

$\xi_x J = A_x$

$V_{i,j,k} = J_{i,j} \Delta\xi \Delta\eta \Delta\zeta$

Metric coefficients × Jacobian represent the area in the physical domain of the faces of the cell ★

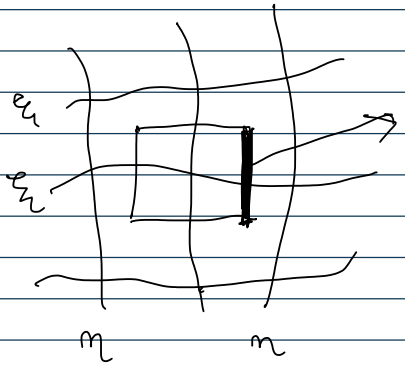
→ This is how you should calculate stuff .... i guess...

(Q2)

Multiplied by J^2 $\quad \frac{\partial^2 \tau}{\partial x^2} + \cdots$

$$\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)\left(\xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta}\right)\tau \quad \cdots$$

$$J^2 \xi_x \frac{\partial}{\partial \xi}\left(\xi_x \frac{\partial \tau}{\partial \xi}\right)$$



$\vec{A_S} = (J\xi_x)\vec{i} + J(\xi_y)\vec{j}$

$\vec{A_S} = A_x \vec{i} + A_y \vec{j}$